

NAPLAN-STYLE PRACTICE - chapter 1

- 1 30 000
- 2 25 004 784
- 3 150
- 4 5
- 5 5
- 6 $1\frac{7}{11}$
- 7 15
- 8 16
- 9 $\frac{2}{15}$
- 10 5
- 11 9.82, 9.209, 8.902, 8.29, 8.2, 8.029
- 12 39.50
- 13 406.737
- 14 10
- 15 \$23.60
- 16 \$6.40
- 17 25.728
- 18 1.25-L bottle for \$3.25 (\$2.60/L)
- 19 $\frac{1}{4}, \frac{3}{5}, \frac{1}{8}, 8.26$
- 20 $0.763\ 491\dots, \frac{2}{3}, \frac{9}{11}$
- 21 $\frac{2}{3}, \frac{9}{11}$
- 22 $0.763\ 491\dots$
- 23 64
- 24 12
- 25 216
- 26 4
- 27 10 cm
- 28 $\frac{5^4 \times 5^8}{5^5 \times 5^7}$
- 29 64
- 30 16
- 31 $\sqrt{3}$

ANALYSIS

- 1 a 1 686 600
- b 7 000 000, 6 000 000, 5 000 000, 2 000 000, 2 000 000, 500 000, 400 000, 200 000
- c 1 000 000; difference is 686 600 people
- d 23 100 000
- e i $\frac{6}{7}$ ii $\frac{1}{10}$ iii $\frac{7}{2}$ or $3\frac{1}{2}$ iv $\frac{5}{6}$
- f i $0.857142 \approx 0.857$ ii 0.1
- iii 3.5 iv $0.83 \approx 0.833$
- g i parts ii, iii ii parts i, iv iii parts i, iv
- a 3 km
- b population = $10\ 000 \times 2^6 = 10\ 000 \times 64 = 640\ 000$
- c approximately 9.22 km
- d Answers to parts a and b are rational numbers as they can be expressed exactly. Answer to part c is an irrational number as it cannot be expressed as an exact fraction or decimal.

NAPLAN-STYLE PRACTICE - chapter 4

- 1 -8
- 2 $2x + y$
- 3 $5 \times a \times b$
- 4 22
- 5 $-3 \times m \times m \times n$
- 6 $6x^2 - x - 4$
- 7 $12ab^2c$
- 8 $\frac{6f}{5}$
- 9 b^6
- 10 $6m^3n^3$
- 11 $\frac{3xy^2}{4}$
- 12 $3w^2$
- 13 $a \times b + a \times c$
- 14 $5x^2 - 10xy$
- 15 1
- 16 $3a(b + 2)$
- 17 $4d + 5d^2$
- 18 $2xy(2x - 3y^2)$
- 19 $5(x - y) + 3(2y - x)$
- 20 $2p(2p - 5q)$
- 21 $14n - m$
- 22 $c + 5$
- 23 length of $(2x + 4)$ m, width of $(3x)$ m
- 24 8 m, 6 m, 48 m²

ANALYSIS

All costs are in dollars.

- a i 30 ii 6
- b i $10x$ ii $2x$
- c i $\frac{3x}{2}$ ii $\frac{6x}{5}$
- d i $16x$ ii $x = 1, 2$ or 3
- e i $10x + 6y$ ii $2(5x + 3y)$
- f i $\frac{2(5x + 3y)}{6} = \frac{5x + 3y}{3}$ ii $\frac{2(5x + 3y)}{p}$
- g Cost of sharing three large boxes and two small boxes of chips between six people is \$7.
- h Some possible answers are: $x = 1, y = 1; x = 1, y = 2; x = 1, y = 3; x = 2, y = 1; x = 3, y = 0; x = 2, y = 0; x = 1, y = 0; x = 0, y = 1; x = 0, y = 2; x = 0, y = 3; x = 0, y = 4; x = 0, y = 5$.
- i Some possible answers are: $x = 1, y = 1, p = 2; x = 1, y = 1, p = 3; x = 3, y = 3, p = 6; x = 3, y = 3, p = 8; x = 4, y = 5, p = 10$.

NAPLAN-STYLE PRACTICE - chapter 2

- 1 C
- 2 36%
- 3 0.42
- 4 0.625
- 5 3.6%, 15%, $\frac{7}{8}, 1.1$
- 6 55%
- 7 87.5%
- 8 \$37.50
- 9 \$578
- 10 \$228
- 11 loss of 15.91%
- 12 125%
- 13 48 minutes
- 14 13:12
- 15 107:65
- 16 30:15:25
- 17 1:2
- 18 5
- 19 $\frac{1}{4}$
- 20 \$110
- 21 \$400
- 22 earning \$15.75 for every hour worked
- 23 12 km/L
- 24 five chocolate bars for \$4.00

ANALYSIS

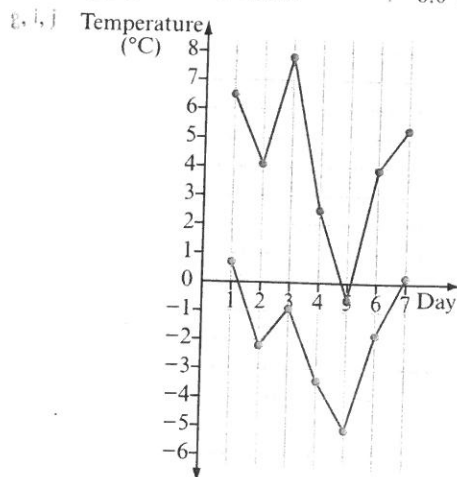
- a i $\frac{13}{20}$ ii 0.65
- b \$13
- c \$33
- d \$455
- e \$3.30
- f \$29.70
- g \$9.70
- h 48.5%
- i 4:5:2
- j day 1: 7 hoodies/h, day 2: 6.875 hoodies/h
- k day 1

NAPLAN-STYLE PRACTICE - chapter 3

- 1 3
- 2 -4.7
- 3 -4
- 4 1
- 5 second level below ground level
- 6 -42
- 7 44
- 8 62 m
- 9 8°C
- 10 -3
- 11 $\frac{6}{1}$
- 12 -4, 7
- 13 -0.6°C
- 14 \$104.90
- 15 $\frac{1}{21}$
- 16 -32
- 17 $(-5)^6$
- 18 -27
- 19 -5
- 20 (0, 0)
- 21 F
- 22 A
- 23 (1, -5)
- 24 (0, -6)
- 25 B and C
- 26 G
- 27 Table 4

ANALYSIS

- a i Wednesday ii Friday
- b 5.8°C
- c Tuesday: 6.3°C, Wednesday: 8.7°C, Thursday: 5.9°C, Friday: 4.5°C, Saturday: 5.7°C, Sunday: 5.1°C
- d Wednesday
- e i -1.8°C ii 4.2°C
- f 6.0°C



- b (1, 0.7), (2, -2.2), (3, -0.9), (4, -3.4), (5, -5.1), (6, -1.8), (7, 0.2)
- c Maximum daily temperatures show similar trend to minimum daily temperatures. Temperature was coldest on Friday before starting to increase again over next two days.

NAPLAN-STYLE PRACTICE - chapter 5

- 1 $x = 25$ 2 $x = 5$
- 3 A number is doubled then 5 is subtracted from this result to give an answer of 25.
- 4 $\frac{n+5}{4} = 1$
- 5 The value of x for the solution must be greater than 15.
- 6 $3r + 15 = 75$ 7 20 8 6
- 9 $x = 5$ 10 $\frac{x}{3} - 8 = 12$ 11 $x = 60$
- 12 $2x = 12$ 13 $4x - 12 = 20$ 14 $x + 8 = 11$
- 15 $x = -3$ 16 $-3x$ 17 $x = -2$
- 18 $y = 3x + 2$ 19 -1 20 $x = 2$

ANALYSIS

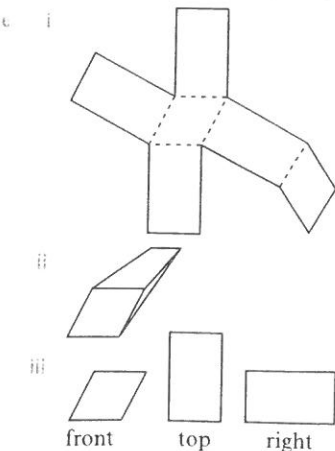
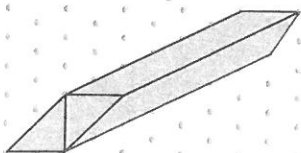
- a $150 + 20x$ b $20x + 150 = 450$
- c $x = 15$; 15 people can attend the party, Kacie can invite 14 friends
- d For 25 people, cost is \$650.
- e no, \$200 over budget
- f $200 + 10x$ g $10x + 200 = 450$; $x = 25$
- h yes i five people j \$250

NAPLAN-STYLE PRACTICE - chapter 7

- 1 isosceles 2 36° 3 kite 4 157°
- 5 3240° 6 C 7 A 8 A
- 9 D 10 D 11 D
- 12 translation of 12 right, 5 down 13 (13, 5)
- 14 rotation, translation, dilation
- 15 SSA 16 B 17 ASA 18 $\frac{1}{3}$

ANALYSIS

- a a reflection about the right side; or a horizontal reflection and a series of translations; or a vertical reflection and a series of translations
- b The front view forms a rhombus, with one pair of opposite angles of 60° and another pair of opposite angles of 120° . Each side length is the same as the original shape side lengths; that is, 2 cm.
- c rhombus-based prism



- f Triangles B and D; these two satisfy the congruency tests (B: SAS and D: ASA). In triangle A, you can't be certain that the side lengths are each 2 cm. In triangle C, you can't be certain that the triangle is equilateral.
- g Only the side lengths are doubled, growing to 4 cm each side. The angle sizes remain at 60° each.

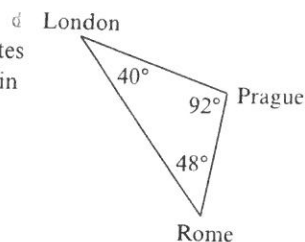
NAPLAN-STYLE PRACTICE - chapter 6

- 1 $\angle DCB$ 2 reflex angle 3 210°
- 4 54° 5 complementary
- 6 $b = 26^\circ$ 7 119°
- 8 Angles are corresponding. 9 125°
- 10 SAS 11 308°T 12 $S65^\circ\text{E}$
- 13 Line of longitude is 153°E . 14 $40^\circ\text{N } 14^\circ\text{E}$
- 15 207 16 15.5 17 6.30 am Sunday
- 18 11.30 pm Friday 19 6.45 pm

ANALYSIS

- i London: $52^\circ\text{N } 0^\circ\text{W}$; Rome: $42^\circ\text{N } 12^\circ\text{E}$; Prague: $50^\circ\text{N } 14^\circ\text{E}$
- ii 133°T iii 010°T iv 285°T

- c 12.25 pm
- d 9 hours and 52 minutes
- f Yes. He arrives back in London at 6.52 pm London time.



6 Connect

NAPLAN-STYLE PRACTICE - chapter 8

- 1 35 2 700 m 3 0.7 m
- 4 14.14 cm 5 20.69 cm 6 125.66 cm
- 7 3.87 cm 8 200 m^2 9 136 m^2
- 10 525 cm^2 11 10 cm 12 1500 cm^2
- 13 14.16 cm^2 14 490.87 cm^2 15 202.63 cm^2
- 16 16 cm 17 686 cm^2 18 949 cm^2
- 19 336 cm^3 20 500 cm^3 21 500 mL
- 22 2000 23 35 cm^2

ANALYSIS

- a 6 m^2 b 10 m c 20 m
- d 40 m^2 for two coats e 825 cm^2
- f 20.73 cm g $34.21\text{ cm}^2 \times 6 = 205.27\text{ cm}^2$
- h 6 m^3
- i $6\text{ m}^3 = 6\,000\,000\text{ cm}^3$; $6\,000\,000 \div 150 = 40\,000$
- j Balls will not take up entire space as there will be space between them. The theoretical calculation does not take into account this space.

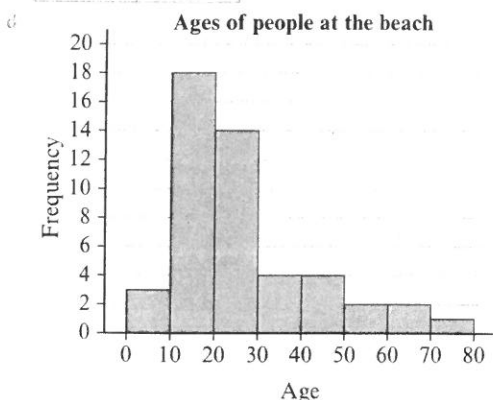
NAPLAN STYLE PRACTICE - chapter 9

- 1 systematic sampling
- 2 asking everyone in your class if the class prefer cats or dogs
- 3 5
- 4 numerical, discrete
- 5 15-19
- 6 F
- 7 39
- 8 scatterplot
- 9 The youngest customer is 13. 10 111 = 1100
- 11 80-100
- 12 roughly symmetrical
- 13 2
- 14 1.7
- 15 4
- 16 2.48, 3, 3, 6
- 17 asking 50 random people from the electoral roll
- 18 mode
- 19 Surveying technique was biased.

ANALYSIS

- a numerical, discrete data
- b systematic sample

Age	Frequency
0-9	3
10-19	18
20-29	14
30-39	4
40-49	4
50-59	2
60-69	2
70-79	1



c Key: 1 | 3 = 13

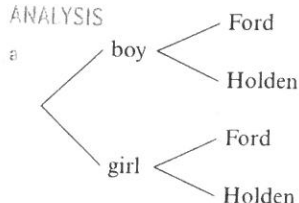
Stem	Leaf
0	5 8 9
1	3 4 5 5 5 6 6 6 6 7 7 8 8 9 9 9 9 9
2	1 1 1 2 2 3 3 3 3 4 5 6 6 7
3	1 2 5 9
4	3 5 6 9
5	6 9
6	1 3
7	2

- f mean $\approx$ 26.7, median = 21.5, mode = 19, range = 67
- g You don't know details such as type of beach, weather on the day, time of year, or if this beach is popular for holiday makers.
- h Results indicate a person chosen randomly at the beach is most likely to be between about 13 and 27.

NAPLAN STYLE PRACTICE - chapter 10

- 1 {yellow, green, blue, pink}
- 2 0.75 or $\frac{3}{4}$
- 3 $\frac{1}{2}$
- 4 $\frac{1}{4}$
- 5 yellow, red or blue
- 6 0.8
- 7 $\frac{8}{20}$ or $\frac{2}{5}$
- 8 $\frac{9}{20}$
- 9 $\frac{7}{9}$
- 10 $\frac{3}{8}$
- 11 $\frac{18}{25}$
- 12 C
- 13 experimental probability = $\frac{11}{30}$; should decrease with further trials
- 14 five
- 15 Table C

ANALYSIS



- b** {BF, BH, GF, GH};

B = boy, G = girl,

F = Ford, H = Holden

Each outcome has theoretical probability of 0.25.

	Ford	Holden	Total
Boys	7	23	30
Girls	19	9	28
Total	26	32	58

58 people were surveyed.

d $\Pr(BF) = \frac{7}{58}$, $\Pr(BH) = \frac{23}{58}$, $\Pr(GF) = \frac{19}{58}$,
 $\Pr(GH) = \frac{9}{58}$

e i $\frac{32}{58}$ ii $\frac{30}{58}$ or $\frac{15}{29}$ iii $\frac{23}{30}$

- f** No; each result would need to have been around 14 or 15.

- g** Possible limitations include age and location of people surveyed.

- h** One possible answer is given.

